The empirical recurrence for OEIS A270511, proved by an exact quasi-polynomial

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Abstract

OEIS A270511 counts triangular arrays of a FIXED shape over the alphabet $0..n$, and carries an empirical recurrence of order 10 contributed by R. H. Hardin. It is true. The array family usually met in this part of the OEIS fixes the alphabet and lets the shape grow, so the count is a walk count in a finite digraph; here it is the alphabet that grows and there is no digraph to walk. Inclusion–exclusion over the adjacent pairs, together with the observation that a set of pairs forces the values along each of its connected components to alternate, gives $a(n)$ in closed form as a quasi-polynomial of period 2, exactly and for every n. The recurrence, and the entry's two branch formulas with it, then follow by polynomial division.

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1 The conjecture

OEIS A270511 is "Number of 3X3X3 triangular 0..n arrays with some element plus some adjacent element totalling n+1 exactly once.". It has offset 1 and begins

$$15, 144, 1137, 4584, 15843, 40392, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-1) +3*a(n-2) -8*a(n-3) -2*a(n-4) +12*a(n-5) -2*a(n-6) -8*a(n-7) +3*a(n-8) +2*a(n-9) -a(n-10)

As of the "Last modified" line on the live entry (Mar 18 2016 08:06:26, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array is the triangle with 3 rows: cells (i, j) for $0 \le i < 3$ and $0 \le j \le i$, so $T = 6$ of them. Two cells are adjacent when they are neighbours in that triangle — side by side in a row, or one directly below the other on either side — and there are $3\binom{3}{2} = 9$ adjacent pairs. Each cell carries a value in $\{0, 1, \ldots, n\}$, and the entry counts those arrays in which EXACTLY ONE adjacent pair $\{u, v\}$ has $x_u + x_v = n + 1$.

Note which parameter grows. The shape is fixed; the alphabet is what n counts. So this is not a walk in a finite digraph, and nothing about transfer matrices applies.

3 A set of pairs forces the values along its components

Write E for the set of adjacent pairs, $t = n + 1$ for the target, and for $S \subseteq E$ let $N(S)$ be the number of arrays satisfying $x_u + x_v = t$ for every pair in S — and possibly others.

Lemma 1. *Let $S \subseteq E$ and let the graph (V, S) have c_b bipartite and c_o non-bipartite components among the cells it touches, $v(S)$ cells being touched. Then*

$$N(S) \;=\; (n+1)^{T-v(S)}\, B^{c_b}\, F^{c_o},$$

where $B = \#\{c : 0 \le c \le n,\ 0 \le t-c \le n\}$ and $F = 1$ if t is even with $0 \le t/2 \le n$, and $F = 0$ otherwise.

Proof. A pair in S forces $x_v = t - x_u$, so within one connected component of (V, S) the value of any one cell determines every other: cells at even distance from a chosen root carry the root's value c, cells at odd distance carry $t - c$. That assignment is consistent exactly when no cycle of the component has odd length; if some cycle does, following it around forces $c = t - c$, so $2c = t$, and the component admits one value when t is even and $t/2$ lies in the alphabet, and none otherwise. A bipartite component is therefore free to choose c subject only to c and $t - c$ both lying in $\{0, \dots, n\}$, which is B choices. Cells touched by no pair of S are unconstrained, giving $(n + 1)$ choices each. $\square$

Here $B = n$ and $F = 1$ precisely when n is odd.

4 The count

Theorem 2. *For every $n \ge 0$,*

$$a(n) \;=\; A(n) + (-1)^n\, C(n), \qquad A(n) = 9n^5 - 36n^4 + 102n^3 - \tfrac{267}{2}n^2 + 93n - 21,$$
$$C(n) = -6n^3 + \tfrac{45}{2}n^2 - 39n + 21.$$

Proof. For an array x let $\sigma(x)$ be its number of satisfied pairs. Then

$$\sum_{S \subseteq E} w^{|S|}\, \#\{x :\ S \subseteq \text{satisfied pairs of } x\} = \sum_x (1+w)^{\sigma(x)},$$

and differentiating in w at $w = -1$ picks out exactly the arrays with $\sigma(x) = 1$:

$$a(n) \;=\; \sum_{\emptyset \ne S \subseteq E} (-1)^{|S|-1}\, |S|\, N(S).$$

Each $N(S)$ is given by Lemma 1. The exponents $T - v(S)$, c_b and c_o do not involve n, so the sum was formed once by running over all 2^9 subsets and recording those three numbers. Every term is a polynomial in n multiplied by F^{c_o}, and F depends only on the parity of n; collecting the two parities gives a polynomial identity in n on each, and the displayed A and C are their half-sum and half-difference. The identity holds for every $n \ge 0$, with no threshold: Lemma 1 is exact at every n. $\square$

The degree of A is 5 and that of C is 3. Both are at most $T - 1$, because a set S with a single pair leaves $T - 2$ cells free and contributes one bipartite component.

5 The two branch formulas

The entry also records, as separate empirical observations:

> Empirical for n mod 2 = 0: a(n) = 9*n^5 - 36*n^4 + 96*n^3 - 111*n^2 + 54*n
>
> Empirical for n mod 2 = 1: a(n) = 9*n^5 - 36*n^4 + 108*n^3 - 156*n^2 + 132*n - 42

These follow at once, and are exact: putting n even and n odd into Theorem 2,

$$a(n) = \begin{cases} 9n^5 - 36n^4 + 96n^3 - 111n^2 + 54n, & n \text{ even}, \\ 9n^5 - 36n^4 + 108n^3 - 156n^2 + 132n - 42, & n \text{ odd}, \end{cases}$$

which is what the entry states.

6 The recurrence

Theorem 3.

$$a(n) \;=\; 2\,a(n{-}1){+}3\,a(n{-}2){-}8\,a(n{-}3){-}2\,a(n{-}4){+}12\,a(n{-}5){-}2\,a(n{-}6){-}8\,a(n{-}7){+}3\,a(n{-}8){+}2\,a(n{-}9){-}a(n{-}$$

for every $n > 10$.

Proof. Write E for the shift operator, $(Ea)(n) = a(n{+}1)$. A polynomial of degree d is annihilated by $(E-1)^{d+1}$ and by nothing smaller, and $(-1)^n$ times a polynomial of degree d is annihilated by $(E+1)^{d+1}$ and by nothing smaller. By Theorem 2 the sequence is therefore annihilated by

$$(E-1)^6(E+1)^4,$$

a monic operator of order 10, and by no proper divisor of it, since the two factors act on the two parts independently and neither part is annihilated by less. The characteristic polynomial of the conjectured recurrence is $t^{10} - \sum_i c_i t^{10-i}$; dividing it by the operator above in $\mathbb{Z}[t]$ leaves no remainder, so the conjectured recurrence is a consequence of Theorem 2 and holds wherever every index it names is at least the offset. □

7 Verification

Three checks, each independent of the algebra above.

First, the closed form reproduces all 27 terms the entry publishes, exactly, in integer arithmetic.

Second, the arrays themselves were enumerated for the smallest alphabets — every assignment of $0..n$ to the 6 cells written out and its satisfied pairs counted, with no inclusion–exclusion and no closed form — and the counts agree. A wrong reading of the cell set, of the adjacency, or of the target would show here and nowhere else.

Third, the conjectured recurrence was applied directly to the closed form's own values over a long range, in exact integer arithmetic, and holds throughout; this is independent of the divisibility argument, which is what actually proves it.

References

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